

Problem #2: Word Embeddings

Solution:

Assignment 1: Match the English words to their Santali translations. (Write down the English words next to the Santali word letters)

<i>Santali word</i>	<i>English word</i>	<i>Santali word</i>	<i>English word</i>	<i>Santali word</i>	<i>English word</i>
A	Mathematics	G	Second	M	Literature
B	Mathematician	H	Position	N	woman
C	Number	I	Time	O	man
D	One	J	Queen	P	girl
E	Two	K	Language	Q	boy
F	First	L	Grammar		

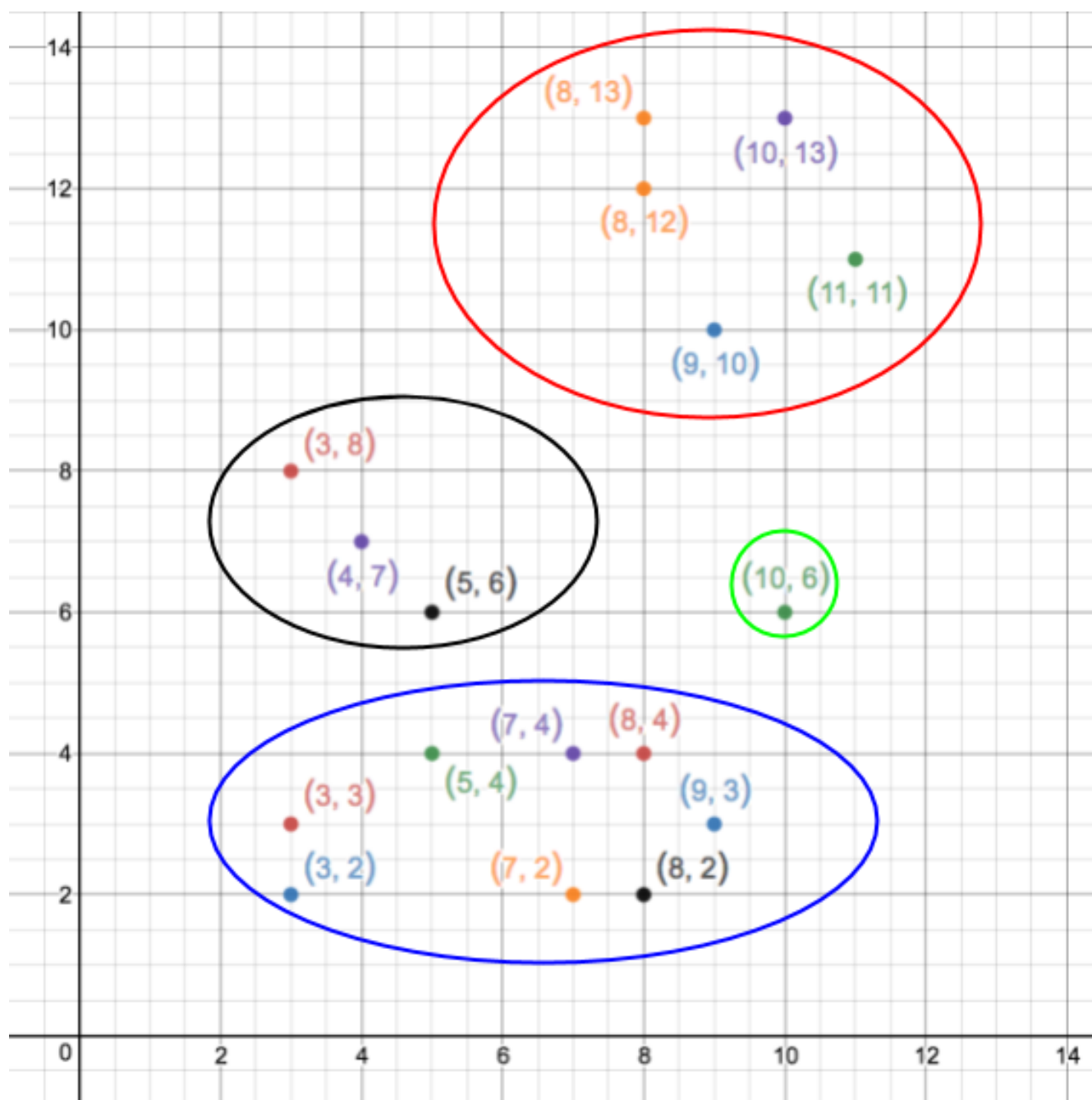
Assignment 2: Guess the embedding of the words

king	<10,14>	linguist	<4,6>	princess	<7, 15>
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Assignment 3: Give English translations of the Santali word. (Anything that means one who studies grammar should be fine).

Grammarians (one who studies grammar)

On a graph, the coordinates appear as follows:



On the graph, we can see that there are clusters of points, and by observing these points we can guess that the cluster circled in blue contains {mathematics, mathematician, number, first, second, one, two, position}, the cluster in red contains {man, woman, boy, girl, queen}, the cluster in black contains {language, literature, grammar} and the cluster in green contains {time}.

We can then use the relations between the English meanings of words, such as:

- *position* has a meaning that is close to *first* and *second*,
- *number* is a feature of the *grammar* of a language
- *second* has a meaning that is close to *time*.

and the similarities between the words in the Santal script (between N and O, P and Q, A and B, etc.) to arrive at the meanings of the words.

For Assignment 2, we can observe the relationships between the points on the graph and see that if we consider each word to be represented by the vectors, then:

- $man - boy = woman - girl \Rightarrow man = boy + woman - girl$
- $two - second = one - first \Rightarrow two = second + one - first$

These vector equations of the form $A = B - C + D$ capture the relation 'A is to B what D is to C.' Similar linear relation can be formulated for the three new words in assignment 2, using which their coordinates can be computed.

- $king = queen + man - woman$
- $linguist = language + mathematician - mathematics$
- $princess = girl + queen - woman$

In Assignment 3, we see that the word is formed from the word for *grammar*, with a suffix that indicates occupation, so we conclude that the embedding for this word should be related to the embedding for *grammar* the same way the embedding for *mathematician* is related to the embedding for *mathematics*.